Chap 8 Quadratic Reciprocity

Q.1

(c) Solve $f(x) = 5x^2 + 6x + 1 \equiv 0 \pmod{23}$

Use the 'quadratic formula' for congruences. The solutions are given by $ax^2 + bx + c \equiv 0$

$$x \equiv (-b + u)(2a)^{-1}$$

where u is a cong-class satisfying

$$u^2 \equiv b^2 - 4ac$$

provided this exists, i.e. that $b^2 - 4ac$ is a quadratic residue.

For this f

$$\begin{aligned} b^2 - 4ac &\equiv 36 - 4.5.1 \pmod{23} \\ &\equiv 36 - 20 \\ &\equiv 16 \\ &\equiv 4^2, 19^2 \end{aligned}$$

So the solutions should be

$$x \equiv (-6 + 4)(10)^{-1}, \quad (-6 + 19)10^{-1} \pmod{23}.$$

$$10^{-1} \equiv 7 \pmod{23}$$

since $7 \cdot 10 = 70 \equiv 1 \pmod{23}$

So our solutions are.

$$x \equiv -2 \cdot 7,\ 13 \cdot 7.$$
$$\equiv -14,\ 91 \pmod{23}$$
$$\equiv 9,\ 22$$

Q9. Let's do instead

Use Euler's criterion to

$$(n|p) \equiv n^{\frac{p-1}{2}} \pmod{p}$$

eg.

$$(8|13) \equiv 8^6 \pmod{13}.$$

Use repeated squaring.

$$6 = 2^1 + 2^2$$
$$8^6 = 8^{2+2^2} = 8^2 \cdot (8^2)^2$$

$$8^2 \equiv 64 \equiv 12 \pmod{13}$$

$$(8^2)^2 \equiv (12)^2 \equiv (-1)^2 \equiv 1 \pmod{13}$$

So $8^6 \equiv 12 \cdot 1 \pmod{13}$

$\equiv -1 \pmod{13}$

$\Rightarrow$ by Euler's Criterion

$(8|13) = -1$, so 8 is

not a quadratic residue mod 13.

Q4. Let p be an odd prime, i.e. $p > 2$.

$\boxed{\text{Let } n \text{ be a quadratic residue mod } p.}$

Prove that n is <u>not</u> a generator

of $\mathbb{Z}_p^\times = U(p)$, ~~the~~ group of

non-zero cong. classes modulo p

under multiplication.

Think about

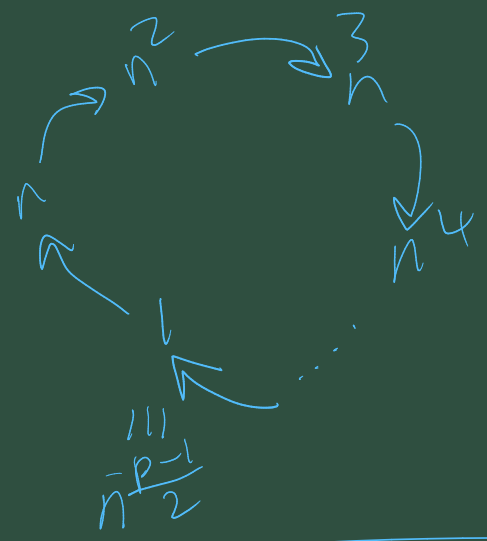
$$\langle n \rangle = \{1, n, n^2, n^3, \ldots\ldots n^{p-1}\}$$

We know $(n|p) = +1$

So Euler's criterion would tell us.

$$(n|p) \equiv n^{\frac{p-1}{2}} \equiv 1 \pmod{p}$$

$$\Rightarrow |\langle n \rangle| \leq \frac{p-1}{2}.$$



$\Rightarrow \langle n \rangle \subsetneq U(p)$.

So n is <u>not</u> a generator of $U(p)$.

<u>OR</u> Since n is a quad-residue, then $n \equiv m^2 \pmod{p}$ for some $m \in U(p)$

$$\langle n \rangle = \{1, n, n^2, n^3, \dots\}$$
$$= \{1, m^2, m^4, m^6, \dots\}$$
$$= \{1, m^2, (m^2)^2, (m^2)^3, \dots, (m^2)^{\frac{p-1}{2}}\}$$

$$(m^2)^{\frac{p-1}{2}} \equiv m^{p-1} \equiv 1.$$

$$\Rightarrow |\langle n \rangle| \leq \frac{p-1}{2}.$$

<u>Q5</u> If p is prime and $p = 2^k + 1$ and n is a non-residue then:

$\langle n \rangle = U(p)$

$|U(p)| = p-1 = 2^k$

~~Consider~~ We want to prove $|\langle n \rangle| = p-1$

or $|n|$ = order of n as an element of group $U(p)$

$= p-1 = 2^k$.

Lagrange's theorem say $|n|$ divides $|U(p)|$

i.e. $|n|$ divides 2^k

$\Rightarrow |n| = 2^l$, for $0 \le l \le k$.

Since n is a quad non-residue, Euler's criterion implies that

$$-1 = (n|p) \equiv n^{\frac{p-1}{2}} \pmod p$$

$$\equiv n^{2^{k-1}}$$

So $n^{2^{k-1}} \equiv -1 \pmod p$

So if $\boxed{|n| = 2^l, \text{ for } l < k}$

then $n^{2^l} \equiv 1 \pmod p$

$$\Rightarrow n^{2^{k-1}} = \left(n^{2^l}\right)^{2^{k-1-l}}, \quad k-1-l \geq 0$$

$$\equiv (1)^{2^{k-1-l}}$$

$\equiv 1$

So this contradicts the fact that n is <u>not</u> a quadratic residue.

Therefore $|n| = 2^k = |U(p)|$

$\Rightarrow U(p) = \langle n \rangle$.

<u>Q11</u>. Evaluate. $(-219 \mid 383)$.

i.e. determine whether -219 is a quad residue or non-residue modulo 383.

QR. $(p \mid q) = \begin{cases} -(q \mid p) & \text{if } p \equiv q \equiv 3 \bmod 4. \\ (q \mid p) & \text{otherwise.} \end{cases}$

$$(-219 \mid 383) = (-1 \mid 383)(219 \mid 383)$$
$$= -(219 \mid 383)$$
$$= -(3 \cdot 73 \mid 383)$$
$$= -(3 \mid 383)(73 \mid 383)$$
$$= -(-(383 \mid 3))(383 \mid 73), \quad \text{by LQR}$$
$$= (383 \mid 3)(383 \mid 73)$$
$$= (2 \mid 3)(18 \mid 73)$$
$$= -(2 \cdot 3^2 \mid 73), \quad \text{since } 2 \text{ is } \underline{\text{not}} \text{ a quadratic modulo } 3.$$

$= -(2|73)(3|73)^2$

$= -(2|73)$, since $(\pm)^2 = +1$.

$= -1$, since $(2|73) = +1$ as $73 \equiv 1 \bmod 8$ by theorem 8.6.

Therefore -219 is <u>not</u> a quadratic residue modulo 383

Q3 Exercise 8.5.

In fact in any arithmetic sequence

$$\{an+b\}_{n=0}^{\infty} = b, b+a, b+2a, b+3a, \ldots$$

provided $\gcd(a,b) = 1$

this sequence will contain an infinite number of primes.

"Dirichlet's Theorem".

General proof needs more

theory. But we can approach some special cases with a <u>similar</u> ideas to Euclid's proof.

a). Claim: There are an infinite number of primes of the form

$$p = 4n-1 \quad / \quad 4m+3$$

3, 7, 11, 15, 19, 23, 27, 31, 35, 39, 43, 47, 51, 55, ...

Proof by contradiction

$$\boxed{\text{Suppose } p \text{ is the largest such prime. so } p = 4N-1}$$

Consider.

$$M = (2^2 \cdot 3 \cdot 5 \cdot 7 \cdot \ldots \cdot p) - 1$$

i.e. (the product of all the primes up to p with an extra 2.) —

Note that $M = 4 \cdot (3 \cdot 5 \cdot \cdots \cdot p) - 1$

So $M > p$, but by the Fundamental Theorem of Arithmetic it has a prime factor q. (M is odd) $q \neq 2$.

Now

$$q \equiv 1 \text{ or } 3 \pmod 4$$
$$\equiv 1 \text{ or } -1 \pmod 4.$$

But if all the prime factors of M were $\equiv 1 \mod 4$. then $M \equiv 1 \pmod 4$

So M has at least one prime factor $q \equiv -1 \pmod 4$

$\Rightarrow$ q is a prime of the form $4a - 1$ for some a.

$\Rightarrow$ q is on the list above and $q \leq p$.

But now:

$q \mid M$ and $M = (2^2 \cdot 3 \cdot 5 \cdots p) - 1$

$$\Rightarrow 1 = (2^2 \cdot 3 \cdot 5 \cdots p) - M$$

$$\Rightarrow q \mid 1, \text{ since } q \mid M \text{ and } q \mid (2^2 \cdot 3 \cdots p)$$

But this contradicts the fact that $q > 1$ as it is prime.

So there is no largest prime of the form $4n-1$ / $4n+3$.

So $\{4n-1 : n \geq 1\}$ contains an infinite number of primes.